

Functions and representations

Name: _____

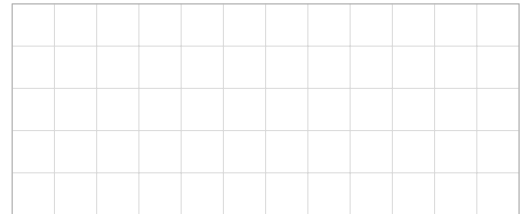
Date: _____

Show your thinking. After checking your work, use the review notes to choose what to revisit in the textbook.

1. A tank starts with 4 liters of water. Water enters at 2 liters per minute for 5 minutes. Let $V(t)$ be the volume t minutes after filling begins.

- (a) Make a table for $t = 0, 1, 2, 3, 4, 5$, and write a formula for $V(t)$.
- (b) Sketch the graph. Label both axes and include units.
- (c) Find $V(3)$ and explain its meaning. State the domain and range for this situation.

After checking: If needed, review §1.1 (function notation, representations, domain, and range) and §1.2 (linear functions).



2. Let $f(x) = x^2 - 3x$.

- (a) Find $f(-2)$ and expand $f(a + h)$.
- (b) Simplify $\frac{f(a + h) - f(a)}{h}$ for $h \neq 0$. Show the cancellation step.

After checking: If needed, review §1.1 (evaluating functions); also revisit expanding binomials and factoring common factors. This algebra will be used in §3.1.

3. Let $g(x) = x + 2$ and $h(x) = x^2$. Find $g(h(x))$ and $h(g(x))$. Explain the order of operations in each composition.

After checking: If needed, review §1.1 (composition of functions).

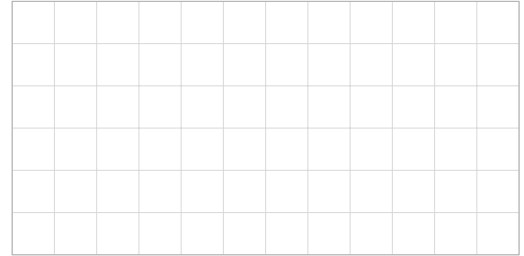
Graphs and domain

4. Consider the function

$$p(x) = \begin{cases} x + 1, & x < 1, \\ 3, & x = 1, \\ 4 - x, & x > 1. \end{cases}$$

- (a) Find $p(0)$, $p(1)$, and $p(2)$.
 (b) Sketch the graph. Mark open and closed endpoints clearly.

After checking: If needed, review §1.2 (piecewise-defined functions) and §1.1 (function notation and graphs).



5. Find the real domain of each function. Use interval notation and explain each restriction.

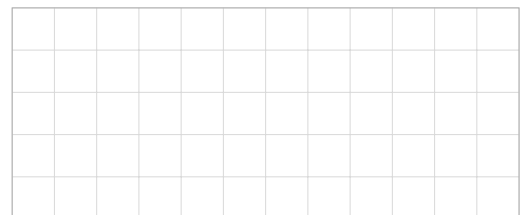
$$(a) g(x) = \sqrt{x - 2} \quad (b) h(x) = \frac{1}{x^2 - 9} \quad (c) k(x) = \ln(x + 1)$$

After checking: If needed, review §1.1 (domain), §1.2 (root and rational functions), and §1.5 (logarithmic functions).

6. A student writes $\frac{x^2 - 4}{x - 2} = x + 2$ for every real x .

- (a) Explain what needs correcting.
 (b) Sketch $y = \frac{x^2 - 4}{x - 2}$, showing any excluded point.
 (c) Do the two formulas define functions with the same domain? Explain.

After checking: If needed, review §1.1 (domain and graphs) and §1.2 (rational functions).



Slope and trigonometry

7. An object's position is $s(t) = t^2$ meters, where t is time in seconds.

- (a) Find the slope of the line through the points on the graph at $t = 1$ and $t = 2$. Interpret the slope, including units.
- (b) Repeat for $t = 1$ and $t = 1.5$. Compare the slopes.
- (c) Write an equation of the line from part (a), using t and s as the coordinates.

After checking: If needed, review §1.2 (slope and equations of lines). These skills will be used for secant lines in §2.1.

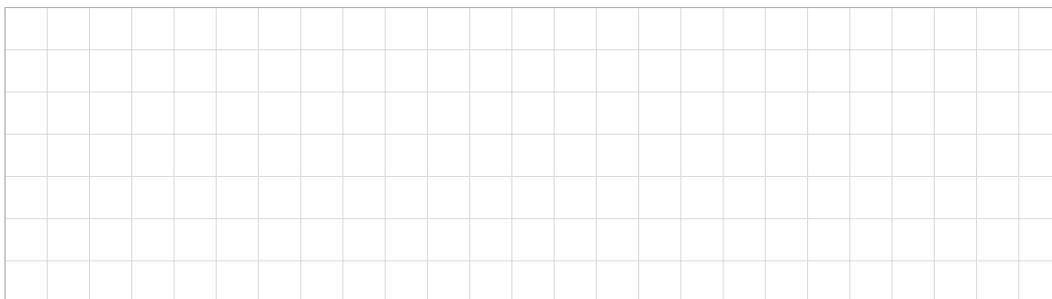
8. For each angle, give its measure in degrees and the coordinates of its point on the unit circle. You may use a unit-circle reference.

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{2}$ (c) π

Explain how the coordinates give the sine and cosine of an angle. *After checking:* If needed, review §1.3 (radian measure and the unit-circle definitions).

9. Sketch $y = \sin x$ for $0 \leq x \leq 2\pi$. Label the intercepts, maximum, and minimum. On the same axes, sketch $y = 2 \sin x$. State the amplitude and period of each graph.

After checking: If needed, review §1.3 (sine graphs, amplitude, and period) and §1.2 (vertical scaling).



Inverses, exponentials, and logarithms

10. Let $f(x) = 3x - 6$.

- (a) Find $f^{-1}(x)$.
- (b) Verify your answer by simplifying $f(f^{-1}(x))$.
- (c) Explain the difference between $f^{-1}(x)$ and $1/f(x)$.

After checking: If needed, review §1.4 (inverse functions and checking inverses).

11. Consider $p(x) = x^2$ with domain all real numbers.

- (a) Explain why p does not have an inverse function on this domain.
- (b) Choose a restricted domain on which p has an inverse. State that inverse and its domain.

After checking: If needed, review §1.4 (one-to-one functions, the horizontal line test, and restricted domains).

12. Work without a calculator.

- (a) Rewrite $2^3 = 8$ in logarithmic form.
- (b) Evaluate $\log_3 9$ and $\ln(e^2)$.
- (c) Solve $e^x = 7$ exactly.
- (d) Are $\ln(a + b)$ and $\ln a + \ln b$ always equal for positive a and b ? Explain using an example.

After checking: If needed, review §1.5 (exponential/logarithmic relationships and logarithm properties).